

AEE Hackathon

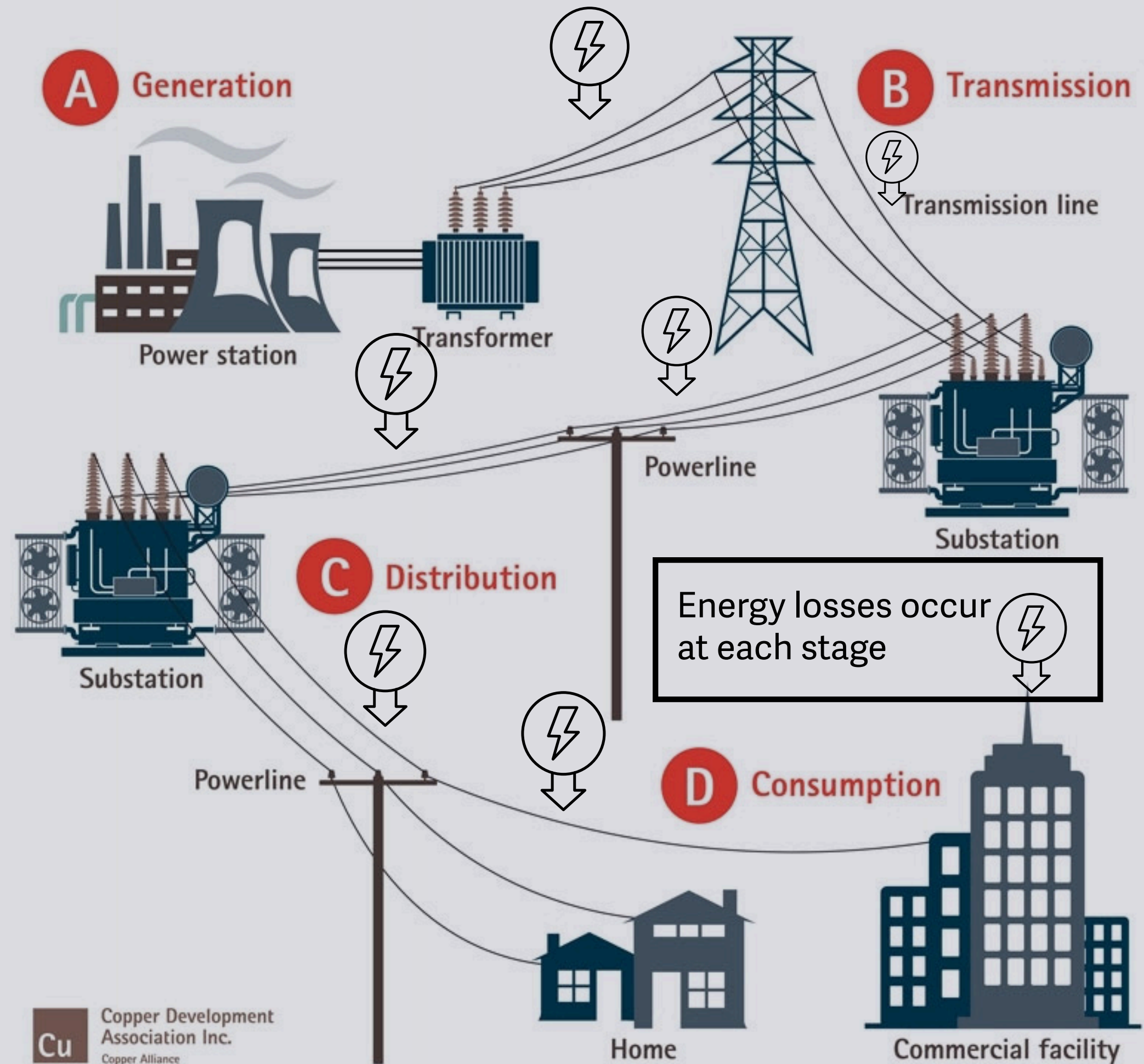
April 19th, 2026

800V BESS DC-DC Converter

Presented by: Reinert Webb

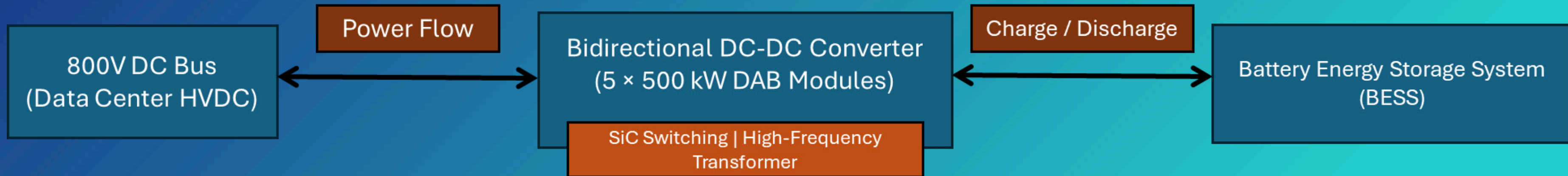
The Challenge

- AI data centers demand massive power
- Multiple conversions reduce efficiency
- Battery storage needed for peak demand
- Poor integration with high-voltage DC



PROPOSED SYSTEM ARCHITECTURE

PROPOSED SOLUTION



- Reduces conversion stages
- Enables bidirectional energy flow
- Improves system efficiency

Current System

- Grid (AC)
- Transformer
- Rectifier (AC > DC)
- UPS system
- Inverter (DC > AC)
- Server power supplies (AC > DC AGAIN)
- Battery (connected through UPS)

Technical Design Choices

[Topology]

Dual Active Bridge (DAB)

- Enables bidirectional power flow
- Provides galvanic isolation via HF transformer
- Supports soft-switching (ZVS) >> lower switching loss
 - Scales well to MW-level modular systems

[Magnetics]

High-Frequency Transformer (Nanocrystalline Core)

- Provides isolation + voltage matching (800V <-> 1000V)
 - Nanocrystalline core > low core loss at high power
 - Copper + core losses included in model
 - Turns ratio ≈ 1.25 supports operating range

[Semiconductors]

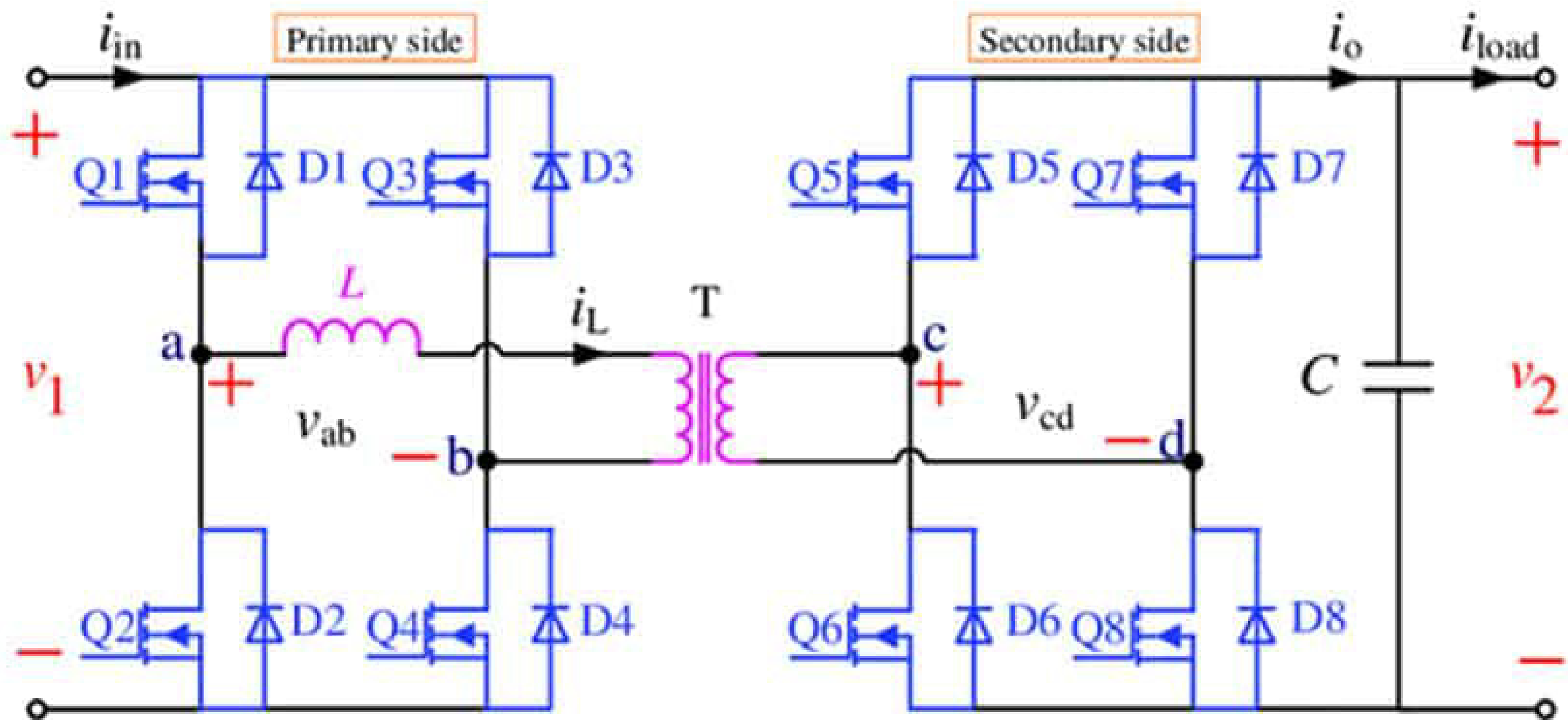
SiC MOSFETs (1700 V class)

- Lower switching losses vs silicon IGBT
- Higher voltage capability > fewer devices
- Enables high-frequency operation (20 kHz)
 - Critical for achieving >99.5% efficiency

[Control Strategy]

Phase-Shift Control (DAB)

- Controls power via phase shift between bridges
 - Supports bidirectional operation
 - Enables CC, CV, CP modes
- Maintains stable power transfer across load range



DC $\rightleftharpoons$ H-Bridge $\rightleftharpoons$ Transformer $\rightleftharpoons$ H-Bridge $\rightleftharpoons$ DC

Results & Requirement Compliance

<u>Requirement</u>	<u>Target</u>	<u>Result</u>	<u>Status</u>
<u>Efficiency @100%</u>	<u>>=99.5%</u>	<u>99.50%</u>	<u>Pass</u>
<u>Efficiency 50-100%</u>	<u>>=99.0%</u>	<u>99.21-99.50%</u>	<u>Pass</u>
<u>Efficiency 25-50%</u>	<u>>=98.5%</u>	<u>98.74-99.21%</u>	<u>Pass</u>
<u>Rated Power</u>	<u>2.5 MW</u>	<u>Achieved</u>	<u>Pass</u>
<u>Voltage Range</u>	<u>720-880 V</u>	<u>Supported</u>	<u>Pass</u>
<u>Isolation</u>	<u>Required</u>	<u>Included</u>	<u>Pass</u>

99.50% Efficiency
at 2.5 MW

